Homework

Homework 9 — Forming a committee — 45 min, after S35.

A team consists of five teachers and seven students, all distinct. A committee of four people is formed.

1. How many different committees can be formed?
2. How many committees contain exactly two students?
3. How many committees contain at least one teacher and at least one student?
4. Among the members of a committee from question 2, a teacher is appointed chair and a student secretary. Count the choices of the committee together with these two appointments.
5. By counting the committees according to their number of teachers, prove that

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4}.$$